

Sensory Nerves

The trigeminal nerve (cranial nerve V) provides the majority of the sensory innervation of the face. It exits the skull via three foramina located bilaterally in the mid-pupillary line: the supraorbital, infraorbital, and mental foramina.

The first branch, the ophthalmic nerve (V_1), has several branches that supply the superior portion of the face: the supraorbital, supratrochlear, infratrochlear, external nasal, and lacrimal nerves. The supraorbital (lateral) and supratrochlear (medial) nerves supply the forehead and anterior scalp and are branches of the frontal nerve—the largest branch of the ophthalmic nerve. These nerves exit through notches along the orbital rim: the supraorbital foramen or notch laterally, and the supratrochlear notch medially. The infratrochlear nerve (a branch of the nasociliary nerve) supplies the glabella, nasal root, and bridge. The external nasal branch (also known as the dorsal nasal nerve), a terminal branch of the anterior ethmoidal nerve from the nasociliary branch of V_1 , innervates the dorsal nose. This provides the anatomical basis for Hutchinson sign in herpes zoster involving the ophthalmic nerve—vesicles on the nasal tip suggest possible ocular involvement because the nasociliary branch sends sensory fibers to both the nasal tip and the cornea. The lacrimal branch of V_1 supplies sensation to the upper eyelid.

The second branch of the trigeminal nerve is the maxillary nerve (V_2). It provides sensory innervation to the lateral nose, lower eyelid, superior cheek, and anterior temple. The maxillary nerve gives rise to two main cutaneous branches: the zygomaticofacial and zygomaticotemporal nerves. The zygomaticofacial nerve exits through the zygomatic bone and supplies a small area around the lateral canthus. The zygomaticotemporal nerve exits through the anterior temporal fossa and innervates the skin of the anterior temporal region. The largest branch of V_2 is the infraorbital nerve, which exits the skull via the infraorbital foramen of the maxilla and supplies sensation to the lower eyelid and superior cheek.

The third branch of the trigeminal nerve is the mandibular nerve (V_3). Its cutaneous branches provide sensory innervation to the lower lip, chin, mandibular and preauricular cheek, anterior ear, and central temporal scalp. The three major cutaneous branches are the auriculotemporal, buccal, and mental nerves. The auriculotemporal nerve innervates the temple, temporoparietal scalp, anterior ear, parts of the external ear canal, and the tympanic membrane. The buccal nerve runs deep to the parotid gland and supplies the skin over the buccinator muscle, buccal mucosa, and gingiva. The mental nerve exits through the mental foramen as a continuation of the inferior alveolar nerve and provides sensation to the lower lip and chin.

Muscles

The scalp contains two muscles: the frontalis anteriorly and the occipitalis posteriorly. These are connected by a tough, fibrous layer called the galea aponeurotica, which spans the cranial vault. The frontalis muscle covers the forehead and elevates the eyebrows. Medial and downward movement of the eyebrows occurs with contraction of the corrugator supercilii muscles. The procerus muscle, located between the corrugator supercilii muscles, pulls the skin of the forehead downward, contributing to horizontal creases at the root of the nose. The orbicularis oculi muscle surrounds the eye and consists of two parts: the orbital portion (involved in forceful eyelid closure) and the palpebral portion (responsible for gentle blinking).